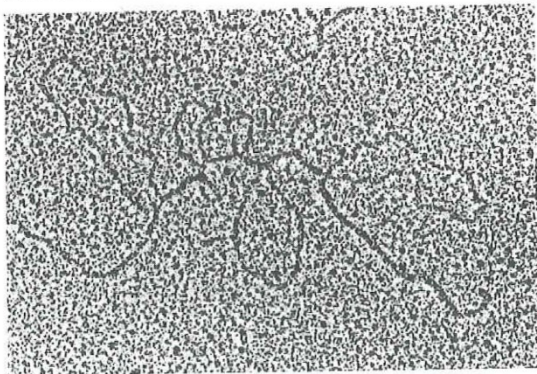


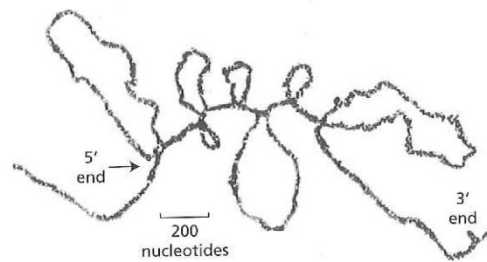
Test question 1

The intron-exon structure of eukaryotic genes came as a shock. In the early, skeptical days, the most convincing demonstration was visual, as shown, for example, by the electron micrograph below (left panel). This image was obtained by hybridizing ovalbumin mRNA to a long segment of DNA that contains the gene. To those used to looking at single- and double-stranded nucleic acids in the electron microscope, the structure was clear: a set of single-stranded tails and loops emanating from a central duplex segment whose ends corresponded to the ends of the mRNA (below, right panel). To the extent possible, describe the intron-exon structure of this gene.

(A) ELECTRON MICROGRAPH



(B) INTERPRETATION



Test question 2

Under stress conditions the levels of the p53 protein regulate cellular decisions about proliferation, apoptosis and senescence.

a) Arginine residues in the interface between p53 and DNA are frequently mutated in tumors. What type of interaction do you expect between an Arg and DNA?

b) What other types of amino acids could make this interaction?

c) A mutation could also affect the ability of p53 to fold into a three-dimensional structure and thereby affect its stability. Different classes of amino acids can cause different folding problems. Please discuss two different ways in which a mutation could cause a problem.

Test question 3

Ribosomes are composed of one large and one small subunit.

In the table below, indicate whether a characteristic is unique to the large subunit, or to the small subunit, or is common for both subunits.

	Large subunit only	Small subunit only	Both subunits
Binds mRNA			
Build of rRNA			
Contains proteins			
Has catalytic activity			
Scans for start-codon			
Interacts with growing peptide			
Binding site for antibiotics			
Binds tRNA			
Slides first during elongation			
Is assembled in the nucle(ol)us			